Prof. Rush's Physics Handout α

[10] **Problem 1.** How long does it take for an object to fall from the lab desk surface to the ground? (Do your best estimation, a range of answers are accepted)

[10] **Problem 2.** In physics, we use **dimensions** to keep track of what kind of units a quantity has. We use L for length, M for mass, and T for time. We can use dimensions for other quantities using these letters. For example:

$$\text{speed } v = \frac{\text{length}}{\text{time}} = \frac{L}{T},$$

$$\text{force } F = \frac{\text{mass} \cdot \text{length}}{\text{time}^2} = \frac{ML}{T^2}.$$

Given speed $= \text{force}^a \cdot \text{mass}^b \cdot \text{time}^c$, or

$$\left(\frac{L}{T}\right) = \left(\frac{ML}{T^2}\right)^a (M)^b (T)^c,$$

find the value of $a + b + c$.

[10] **Problem 3.** Using known properties of the Earth, estimate the gravitational constant G.

Given:

$$M_{earth} = 5.97 \times 10^{24}\,\text{kg}, \quad R_{earth}^2 = 4.057 \times 10^{13}\,\text{m}, \quad g = 9.8\,\text{m/s}^2.$$

Use the relation $g = \dfrac{GM_{earth}}{R_{earth}^2}$ to compute G, and round your answer to 2 significant figures (e.g. 1.2×10^{-3}).

[10] **Problem 4.** A mass m moving at speed v collides elastically with a stationary mass m. After collision, the two masses scatter at equal angles θ with respect to the incident direction (the direction the first mass was initially moving in) Find the measure of angle θ. *Hint:* The momentum, or mass times velocity of the system is the same before and after the collision.

[10] **Problem 5.** If the entire solar system were magnified so that all distances were 67 times larger than their actual values, while the densities of all bodies remained unchanged, determine the new duration of Earth's orbital period (its "year").

Express your answer in **Earth years**.

[12] **Problem 6.** Hina is falling through the sky (assume gravitational constant g everywhere, and assume no air drag), directly below her, there are infinite layers of very dense and thin clouds, vertically separated by uniform distances h. When she falls through each cloud, her speed would be decreased by a small positive quantity Δv. After sufficient time, Hina's speed would follow a periodic pattern. Find her average speed over infinite time.

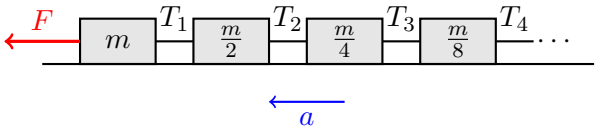
[15] **Problem 7.** N identical uniform bricks, each of length L, are stacked, one above the other, near the edge of a table. What is the maximum possible length the top brick can protrude over the edge of the table? (Express your answer as the sum of a series)

Hint: The stack remains stable only if, for each group of bricks counted from the top down, the center of mass of that group lies directly above the brick (or table) that supports it.

[30] **Problem 8.** An infinite chain of blocks is connected by massless, inextensible strings and moves without friction along a straight horizontal surface. The front block is pulled with a horizontal force F, causing the entire chain to accelerate at a constant rate a. The front block has mass m, and each subsequent block has half the mass of the block directly in front of it, as shown in the figure below.

(a) Find the force F.

(b) Find the tension T_n in the nth string in terms of n, F, a.



[12] **Problem 9.** Suppose you are at the 41st floor of the Empire State Building. You take an elevator to the 67th floor. Circle each correct option (You only have one attempt for these problems!).

(a) At some point in the journey, you will feel lighter. (True/False)

(b) At some point in the journey, you will feel heavier. (True/False)

(c) A bowl of water sits on the floor of the elevator, and a mass floats in it. For incompressible ideal water, the mass will stay in place relative to the bowl for the entire journey. (True/False)

[40] **Problem 10.** A projectile is launched from the base of a straight ramp that makes an angle φ above the horizontal. The launch speed is v, and the launch angle θ (above horizontal) is adjustable. Neglect air resistance and assume uniform gravity g.

(a) Determine the launch angle θ that maximizes the distance along the slope before impact, express it in terms of φ.

(b) Express the maximum range along the slope in terms of v, g, φ.

Hint: You may find it useful to use the auxiliary angle formula

$$a\sin x + b\cos x = \sqrt{a^2+b^2}\,\sin(x+\delta), \qquad \text{where } \tan\delta = \frac{b}{a}.$$